

JomCode Coding Bootcamp

SQLAlchemy PostgreSQL persistence

Understand, predict, type, run, then practice.

What you will be able to do

- 01 Map three relational tables with SQLAlchemy.
- 02 Query through scoped sessions and avoid string-built SQL.
- 03 Distinguish owner schema setup from the restricted runtime connection.

Plain meaning → technical view → use

PLAIN MEANING

Work with database rows using Python objects and safe queries.

TECHNICAL VIEW

SQLAlchemy maps classes to tables and uses sessions, transactions and bound parameters.

WHEN YOU USE IT

Create a workshop or reserve a seat safely in PostgreSQL.

TIP

Use one database session per request and bound values for user input.

Install the persistence packages

TERMINAL >_ PowerShell / zsh

```
> python -m pip install sqlalchemy psycopg[binary]
```

Use a virtual environment. Keep the real database URL private.

Worked example 1: Mapped table names



chapter_16_sqlalchemy_postgresql_persistence.py

```
1 from app.models import User, Workshop, Registration
2 print(User.__table__.schema)
3 print(Workshop.__table__.name)
4 print(Registration.__table__.name)
```

PREDICT Where is the schema declared?

Example 1: output and reason

TERMINAL >_ EXPECTED OUTPUT

```
bootcamp
workshops
registrations
```

WHY THIS HAPPENS

Each SQLAlchemy model declares its table schema.

Exercise 1: Literal title search

BUILD

Build a SQLAlchemy query that treats % as search text rather than a wildcard.

CHECK

A bound parameter exists and contains(autoescape=True) is used.

IF STUCK

Do not build SQL by concatenating user text.

Worked example 2: Bound parameter



chapter_16_sqlalchemy_postgresql_persistence.py

```
1 from sqlalchemy import select
2 from app.models import Workshop
3 query = select(Workshop).where(Workshop.id == "w1")
4 print(query.compile().params)
```

PREDICT Does the ID join the SQL string?

Example 2: output and reason

TERMINAL >_ EXPECTED OUTPUT

```
{'id_1': 'w1'}
```

WHY THIS HAPPENS

No. SQLAlchemy binds the value as a parameter.

Exercise 2: Registration transaction

BUILD

Explain how a row lock, duplicate check, count and insert protect the last seat.

CHECK

Exactly one registration can take one last seat.

IF STUCK

All four operations must share a transaction.

Worked example 3: Row lock intent



chapter_16_sqlalchemy_postgresql_persistence.py

```
1 from sqlalchemy import select
2 from sqlalchemy.dialects import postgresql
3 from app.models import Workshop
4 query = select(Workshop).where(Workshop.id == "w1").with_for_update()
5 print("FOR UPDATE" in str(query.compile(dialect=postgresql.dialect())))
```

PREDICT Does compiling acquire a lock?

Example 3: output and reason

TERMINAL	>_	EXPECTED OUTPUT
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		True
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WHY THIS HAPPENS

No. A database transaction acquires the lock when it executes the query.

Exercise 3: Session scope

BUILD

Explain when the SQLAlchemy session closes and when writes commit.

CHECK

One request gets one scoped session; failed writes roll back.

IF STUCK

A global unclosed session is not a safe shortcut.

Checkpoint

PROVE IT

Learners can map each table, explain bound values, and describe one safe registration transaction.